A

Major Project

On

NLP BASED ANALYSIS OF CYBER THREATS AND VULNERABILITIES OF THE HEALTHCARE ECOSYSTEM

(Submitted in partial fulfilment of the requirements of the award of Degree)

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In

INFORMATION TECHNOLOGY

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May, 2026

DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that the project entitled "**NLP-BASED ANALYSIS OF CYBER THREATS AND VULNERABILITIES OF THE HEALTHCARE ECOSYSTEM**" being submitted by **D. Sruthi (227R1A1281)**, **Md Samiuddin (237R5A1207) & P. Nitheesh (227R1A12C1)** in partial fulfilment of the requirements for the award of the degree of B.Tech in Information Technology to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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Submitted for viva voice Examination held on _____

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ABSTRACT

Digitization in healthcare systems, with the wid adoption of Electronic Health Records, connected medical devices, software and systems providing efficient healthcare service delivery and management. On the other hand, the use of these systems has significantly increased cyber threats in the healthcare sector. Vulnerabilities in the existing and legacy systems are one of the key causes for the threats and related risks. Understanding and addressing the threats from the connected medical devices and other parts of the ICT health infrastructure are of paramount importance for ensuring security within the overall healthcare ecosystem. Threat and vulnerability analysis provides an effective way to lower the impact of risks relating to the existing vulnerabilities. However, this is a challenging task due to the availability of massive data which makes it difficult to identify potential patterns of security issues. This project proposes an NLP-driven framework for automated cyber-threat analysis using machine-learning and transformer-based feature representation. Textual threat reports are preprocessed through tokenization, lemmatization, stop-word removal, and linguistic normalization, after which contextual semantic features are extracted using DeBERTa word embeddings. The extracted feature space is evaluated using multiple machine-learning classifiers, including Greedy Tree Classifier, TaoTree Classifier, K-Nearest Neighbors (KNN), and Gaussian Naive Bayes, enabling comparative performance assessment across traditional and rule-based interpretable learning models. The proposed system further employs the Greedy Tree model as the optimized predictive engine for determining three key outcomes: Threat Category, Severity Score, and Suggested Defense Mechanism. The framework is deployed as a web-based application that supports secure test-data upload, automated inference, and visualization of predicted results to support cybersecurity analysts in decision-making. Experimental evaluation demonstrates improved interpretability, threat prioritization capability, and reliable detection of high-risk attacks. The system enhances proactive cyber-risk management in healthcare by integrating NLP, transformer embeddings, and multi-model machine-learning analytics for intelligent cyber-threat assessment.

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1. INTRODUCTION

1. INTRODUCTION

The rapid digitization of the healthcare sector has significantly improved patient care, operational efficiency, and data management through technologies such as Electronic Health Records (EHRs), connected medical devices, and cloud-based systems. However, this digital transformation has also increased the exposure of healthcare systems to cyber threats such as malware, ransomware, phishing, and Distributed Denial of Service (DDoS) attacks.

To address these challenges, this project introduces an NLP-based framework that leverages advanced machine learning and transformer-based models to automate cyber threat analysis. By preprocessing textual data, extracting contextual features using DeBERTa embeddings, and applying multiple classifiers, the system can accurately classify threats, estimate severity levels, and recommend appropriate defense mechanisms

1.1 PROJECT PURPOSE

Several research works have contributed to the development of cyber threat detection systems using advanced technologies. Researchers have explored various machine learning and deep learning techniques to identify and analyze vulnerabilities. One study presented a survey of machine learning and data mining techniques used to reduce the impact of software vulnerabilities, showing how these methods can improve cybersecurity systems.

This system is capable of identifying important details such as data breaches, ransomware attacks, and phishing incidents. Similarly, researchers have used Named Entity Recognition techniques to identify key entities in cybersecurity text, such as software names, vendors, and versions, using models like BiLSTM-CRF combined with attention mechanisms.

In addition, some researchers have developed systems for detecting vulnerabilities in software by treating source code as text and applying natural language processing techniques. These approaches have achieved high accuracy, demonstrating the effectiveness of deep learning models in cybersecurity applications. Transformer-based models such as BERT and its variants have also been widely used to improve the performance of text analysis tasks in the cybersecurity domain.

1.2 PROJECT FEATURES

The proposed system includes several advanced features that enable efficient and intelligent cyber threat analysis:

1. Natural Language Processing (NLP) Based Preprocessing

The system uses Natural Language Processing techniques to clean and prepare the raw text data for analysis. In real-world scenarios, cyber threat data often contains unnecessary words, symbols, and noise. The preprocessing step removes unwanted elements, breaks the text into smaller parts, and converts it into a structured format. This helps the system understand the data better and improves the overall performance of the model.

2. Advanced Feature Extraction Using Transformer Models

The project uses advanced transformer-based models like DeBERTa to extract meaningful features from the text. These models understand the context and relationships between words rather than just reading them individually. This allows the system to capture deeper meanings in the data, leading to more accurate predictions.

3. Multiple Machine Learning Algorithms

The system uses different machine learning algorithms such as Greedy Tree, TaoTree, K-Nearest Neighbors, and Naive Bayes. Each algorithm has its own strengths, and using multiple models helps in comparing their performance and selecting the best one. This improves the reliability and effectiveness of the system.

4. Cyber Threat Classification

One of the main features of the project is its ability to classify cyber threats into different categories. For example, it can identify whether a threat is related to malware, phishing, ransomware, or other types of attacks. This classification helps users quickly understand the nature of the threat.

5. Severity Level Prediction

The system not only identifies the type of threat but also predicts how serious it is. It assigns a severity level to each threat, which helps users prioritize their response. High-severity threats can be handled immediately, while lower-severity threats can be addressed later.

6. Defense Mechanism Suggestion

Based on the identified threat and its severity, the system suggests possible defense mechanisms. This helps users take appropriate action without needing deep knowledge of cybersecurity. It acts as a guide for handling different types of cyber threats effectively.

7. Web-Based User Interface

The project is implemented as a web application using Django. This provides a simple and user-friendly interface where users can upload datasets, run analysis, and view results. It does not require advanced technical skills, making it easy to use for beginners.

8. Automated Processing and Results Generation

The entire process, from data input to final prediction, is automated. Once the user uploads the data, the system performs all steps including preprocessing, analysis, and prediction automatically. This reduces manual effort and speeds up the process.

2. LITERATURE SURVEY

2. LITERATURE SURVEY

There are several recent works that focus on threat and vulnerability detection and analysis based on Machine Learning (ML) models. In Ghaffarian et al. [1], a survey of ML and Data Mining techniques to mitigate the damages of software vulnerabilities is presented. In Satyapanich et al. [2], a semantic schema to describe CS events was presented using Deep Learning-based Information Extraction (IE) pipeline to implement the automatic extraction of structured information about data breaches, ransomware and phishing attacks and the discovery and the patches of vulnerabilities. In Gao et al. [3], a data and knowledge-driven CS Named Entity Recognition (NER) method is presented, exploiting a Bidirectional Long Short Term Memory with Conditional Random Field (BiLSTM-CRF) architecture, including also a multi-head self-attention neural network with word embeddings trained on CS closed-domain texts to improve their effectiveness in conjunction with KBs, for the recognition of the details of the assets (application, vendor, version, etc.) involved in CS issues. In Nikoloudakis et al. [4], a ML-based situational awareness framework is presented which is able to detect existing and newly introduced network-enabled entities in an IoT-based environment based on real-time awareness features provided by the Software-Defined Networking (SDN) paradigm, assessing them against known vulnerabilities, and assigning them to a connectivity-appropriate network slice. The authors of [5] developed software vulnerability detection as an NLP problem with source code treated as texts, addressing the automated software vulnerability detection using recent DL NLP models. They compared various DL models based on their accuracy and the best performer achieved 95% of accuracy. Furthermore, the proposed approach was also able to predict the vulnerability class of source codes. The authors of [6] presented an NLP DL-based architecture for the identification of relevant CS information, such as vulnerability exploitations, attack discoveries and advanced persistent threats. This architecture is composed of a word-embedding layer, a BiLSTM layer, and a CRF layer, concatenated with a further BiLSTM as output layer. The results of their experiments showed some improvements with respect to the baselines. The authors of one paper [7] presented a method to analyze the severity of CS threats analyzing the language of CS-related tweets through a DL approach. The experiments used a corpus of 6000 tweets containing the description of software vulnerabilities, annotated with the opinions of the authors toward their severity. The paper also presented a method for linking software vulnerabilities reported in tweets to CVEs and NVD KBs. The obtained results demonstrated a high-precision in forecasting high-severity vulnerabilities, also highlighting that reports of severe vulnerabilities extracted from online

sources are predictive of real-world exploits. According to [8], researchers need to deeply investigate ethical compliance even when the data seem to be public. Usually, in CS research the data are accessed and analyzed without the informed consent of participants, but acquiring informed consent could be practically impossible with datasets containing hundreds of data. In the case of the experimental assessment presented in this work, there is no personal data included, so there are no ethical issues. The authors of [9] presented a method for NER in the CS domain that uses a model that integrates BERT and BiLSTM-CRF DL architectures, improving baseline performance. As per recent study showed that at least 20% of the medical device manufacturers experienced ransomware or malware attacks in the last 20 months. The authors of [10] proposed a cyber supply chain threat analysis that integrates Random Forest and XGBoost algorithms for the threat prediction. The work considers threat intelligence and predicts the Tactics, Techniques, and Procedures (TTP) deployed for a cyber attack, demonstrating high accuracy in their experimental assessment.

2.1 REVIEW OF RELATED WORK

Several researchers have contributed to the field of cyber threat analysis using machine learning and NLP techniques.Ghaffarian et al. presented a survey on ML and data mining techniques for mitigating software vulnerabilities, highlighting their effectiveness in reducing security risks. Satyapanich et al. proposed a deep learning-based information extraction pipeline to automatically extract structured information about cyber incidents such as ransomware and phishing attacks.

Gao et al. introduced a Named Entity Recognition (NER) approach using BiLSTM-CRF architecture combined with attention mechanisms to identify important cybersecurity entities like software, vendors, and vulnerabilities. Similarly, Nikoloudakis et al. developed a machine learning-based situational awareness framework for IoT environments to detect vulnerabilities and assign appropriate security measures.

2.2 DEFINITION OF PROBLEM STATEMENT

The healthcare ecosystem generates large volumes of cyber-threat data from logs, incident reports, and security alerts, but traditional systems rely on manual analysis, static rule-based detection, and fragmented monitoring tools, which are slow, error-prone, and unable to detect evolving attacks such as ransomware, phishing, malware, and DDoS. These conventional approaches lack the ability to process unstructured textual threat information, fail to classify attacks by severity or type, and provide limited support for proactive threat prioritization, resulting in delayed response, higher vulnerability exposure, and increased risk to patient data and clinical operations. To overcome these challenges, the proposed system introduces an NLP-based automated framework that preprocesses cyber-threat text, extracts semantic features using DeBERTa embeddings, and applies machine-learning models such as Greedy Tree, TaoTree, KNN, and Gaussian Naive Bayes to classify threat category, estimate severity level, and suggest suitable defense mechanisms, thereby improving accuracy, automation, and security decision-making in healthcare environments.

2.3 EXISTING SYSTEM

The existing system for analyzing cyber threats in the healthcare sector is mainly based on traditional approaches that depend on manual work and predefined rules. In these systems, cybersecurity analysts manually examine large volumes of data to identify potential threats. This process is time-consuming and requires significant human effort.

Traditional systems also use rule-based techniques, where predefined conditions are used to detect threats. For example, if certain keywords or patterns are found in the data, the system generates an alert. However, these rules are static and cannot adapt to new or unknown threats. As cyber-attacks continue to evolve, these systems become less effective.

Another limitation of existing systems is that they rely heavily on keyword matching. They simply look for specific words such as “malware” or “phishing” without understanding the context in which these words are used.

Disadvantages

1. Traditional systems rely on exact keyword or pattern matching (RegEx). They are incapable of understanding synonyms, variations in sentence structure, or novel jargon, leading to failure in classifying new or subtly disguised threats that do not perfectly match the hard-coded dictionary.
2. The process hinges on manual human effort for reading, feature extraction, and rule creation. The sheer volume of modern cyber threat data overwhelms human analysts, leading to significant delays, bottlenecks, and backlogs in threat analysis and response time.
3. As the threat landscape evolves, the number of hard-coded IF-THEN rules required to cover all variations grows exponentially. This leads to an unmanageable "rule explosion," where maintaining, debugging, and resolving conflicts between thousands of individual rules becomes extremely complex and error-prone.
4. The system is inherently reactive. It can only identify threats for which a known signature or rule already exists. It is completely ineffective at classifying or even flagging genuinely new, zero-day vulnerabilities describ
5. ed using fresh, unanticipated language.

2.4 PROPOSED SYSTEM

To overcome the limitations of traditional systems, this project proposes an advanced solution based on natural language processing and machine learning techniques. The proposed system is designed to automatically analyze cyber threat data and provide accurate results.

In this system, the input data is first preprocessed using NLP techniques such as tokenization, stop-word removal, and lemmatization. This step converts raw text into a structured format that can be easily processed by machine learning models. After preprocessing, advanced transformer models like DeBERTa are used to extract meaningful features from the text. These features capture the context and semantic meaning of the data.

The system then uses multiple machine learning algorithms such as Greedy Tree, TaoTree, K-Nearest Neighbors, and Naive Bayes to classify cyber threats. Based on the analysis, the system predicts the type of threat, its severity level, and suggests suitable defense mechanisms.

2.5 OBJECTIVES

- To develop an NLP-based framework for analyzing cyber-threat and vulnerability data within the healthcare ecosystem.
- To preprocess and transform unstructured textual threat reports using tokenization, lemmatization, and contextual feature extraction with DeBERTa embeddings.
- To evaluate multiple machine-learning models — including Greedy Tree, TaoTree, K-Nearest Neighbors (KNN), and Gaussian Naive Bayes — for threat classification and severity prediction.
- To design a predictive system capable of identifying threat category, severity score, and suitable defense mechanism with improved accuracy and interpretability.
- To compare classifier performance using standard evaluation metrics and visualization-based performance analysis.

Advantages of the proposed system

1. Automation of Cyber Threat Analysis

One of the biggest advantages of this project is that it automates the process of analyzing cyber threats. In traditional systems, security experts need to manually read and analyze large amounts of data such as logs and reports, which takes a lot of time and effort. This project reduces that burden by automatically processing the data and generating results. This not only saves time but also increases efficiency and productivity.

2. Ability to Handle Unstructured Data

Most cyber threat information is available in the form of unstructured text, such as security alerts, incident reports, and vulnerability descriptions. Traditional systems struggle to understand such data because they rely on fixed rules. This project uses Natural Language Processing techniques to understand the meaning and context of the text, making it easier to extract useful information from complex data.

3. Improved Accuracy in Threat Detection

The project uses advanced machine learning models and transformer-based techniques, which help in identifying patterns more accurately. Compared to rule-based systems, this approach reduces errors such as false alarms or missed threats. As a result, the system provides more reliable and accurate predictions.

4. Threat Classification and Severity Prediction

Another important advantage is that the system not only detects threats but also classifies them into categories like malware, phishing, or ransomware. In addition to this, it predicts the severity level of each threat. This helps users understand how serious a threat is and allows them to take appropriate actions quickly.

5. Recommendation of Defense Mechanisms

The system provides suggestions on how to handle detected threats. Based on the type and severity of the threat, it recommends suitable defense mechanisms. This is very useful for beginners or organizations that may not have deep cybersecurity expertise, as it guides them on what steps to take next.

6. Scalability and Efficiency

The project is designed in such a way that it can handle large amounts of data without slowing down. As the amount of cyber threat data increases, the system can still perform efficiently. This makes it suitable for real-world applications, especially in healthcare systems where data is continuously generated

2.6 HARDWARE & SOFTWARE REQUIREMENTS

The system requires appropriate hardware and software resources to ensure efficient execution of NLP processing, model training, and prediction tasks.

2.6.1 HARDWARE REQUIREMENTS

- **Processor:** Intel Core i5 or higher (or equivalent AMD Ryzen)
- **RAM:** Minimum 8 GB (16 GB recommended)
- **Storage:** At least 250 GB (SSD preferred)
- **GPU (Optional):** NVIDIA GPU for faster computation
- **Display:** Minimum resolution 1366×76.

2.6.2 SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10/11, Linux (Ubuntu), or macOS
- **Programming Language:** Python 3.x
- **Web Framework:** Django
- **Data Tools:** Pandas, NumPy
- **Visualization:** Matplotlib, Seaborn
- **Database:** MySQL

3. SYSTEM ARCHITECTURE & DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

3.1 PROJECT ARCHITECTURE

The proposed system is an advanced Machine Learning pipeline designed for the NLP-Based Analysis of Cyber Threats and Vulnerabilities in the Healthcare Ecosystem. It starts by preprocessing raw threat text using NLTK for cleaning and lemmatization, while simultaneously encoding the three target variables (Threat Category, Severity Score, Defense Mechanism). The core innovation lies in the use of the DeBERTa Transformer model to generate dense, context-aware embeddings from the text, effectively capturing semantic meaning that traditional systems missed. After balancing the imbalanced dataset using SMOTE, the rich DeBERTa features are fed into multiple supervised classifiers, primarily the Greedy Tree Classifier (Proposed Model), which is chosen for its balance of high predictive accuracy and crucial model interpretability, allowing the system to provide fast, accurate, and explainable threat classifications for actionable defense strategies.

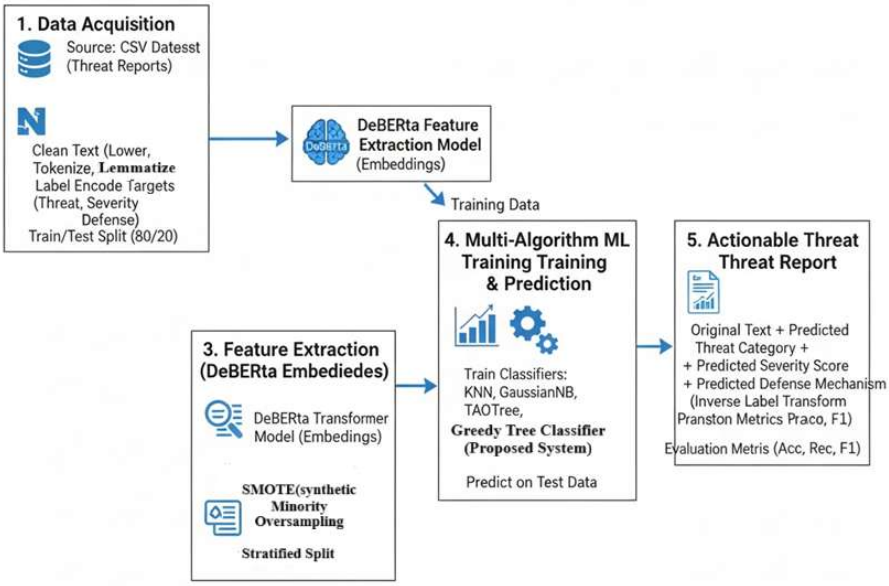


Figure 3.1 Proposed System Architecture of Nlp Based Analysis of Cyber Threats and Vulnerabilities of Healthcare Ecosystem

3.2 DESCRIPTION

1. Data Acquisition and Dataset Loading

The system initiates with the Data Acquisition step, where the core dataset is loaded into a pandas DataFrame using the upload_dataset function. This dataset contains the raw, unstructured text descriptions of cyber threats and the corresponding human-labeled categorical columns: Threat Category, Severity Score, and Suggested Defense Mechanism.

2. Text Preprocessing and Label Encoding

This critical clean_text pipeline involves lower-casing the text, performing tokenization, removing common English stop words, and applying WordNet Lemmatization to reduce words to their dictionary roots

3. Deep Contextual Feature Extraction (DeBERTa Embeddings)

This step leverages the state-of-the-art DeBERTa Transformer model for feature engineering. The preprocessed text is input into DeBERTa, which converts the documents into high-dimensional, dense numerical vectors known as embeddings.

4. Data Balancing and Stratified Train/Test Split

To counteract the inherent class imbalance in cyber security data, a critical balancing technique is applied. The DeBERTa feature vectors are simultaneously oversampled and then subjected to SMOTE which generates synthetic data points for the minority classes, ensuring the training data is balanced for each of the three classification tasks

5. Multi-Algorithm Model Training and Selection

Multiple machine learning classifiers, including KNN, Gaussian NB, and two Interpretable Decision Trees (TAO Tree and Greedy Tree Classifier), are trained independently for each of the three targets.

3.3 UML DIAGRAM

3.3.1 Class Diagram

Class diagram represents the core structure of the NLP-based cyber-threat analysis system, highlighting major modules such as preprocessing, feature extraction, model inference, metrics evaluation, authentication, and database connectivity. It shows how textual threat data flows from preprocessing to DeBERTa-based feature extraction, passes through multiple ML algorithms including Greedy Tree, TaoTree, KNN, and Gaussian Naive Bayes, and finally reaches the prediction and evaluation layers for actionable output.

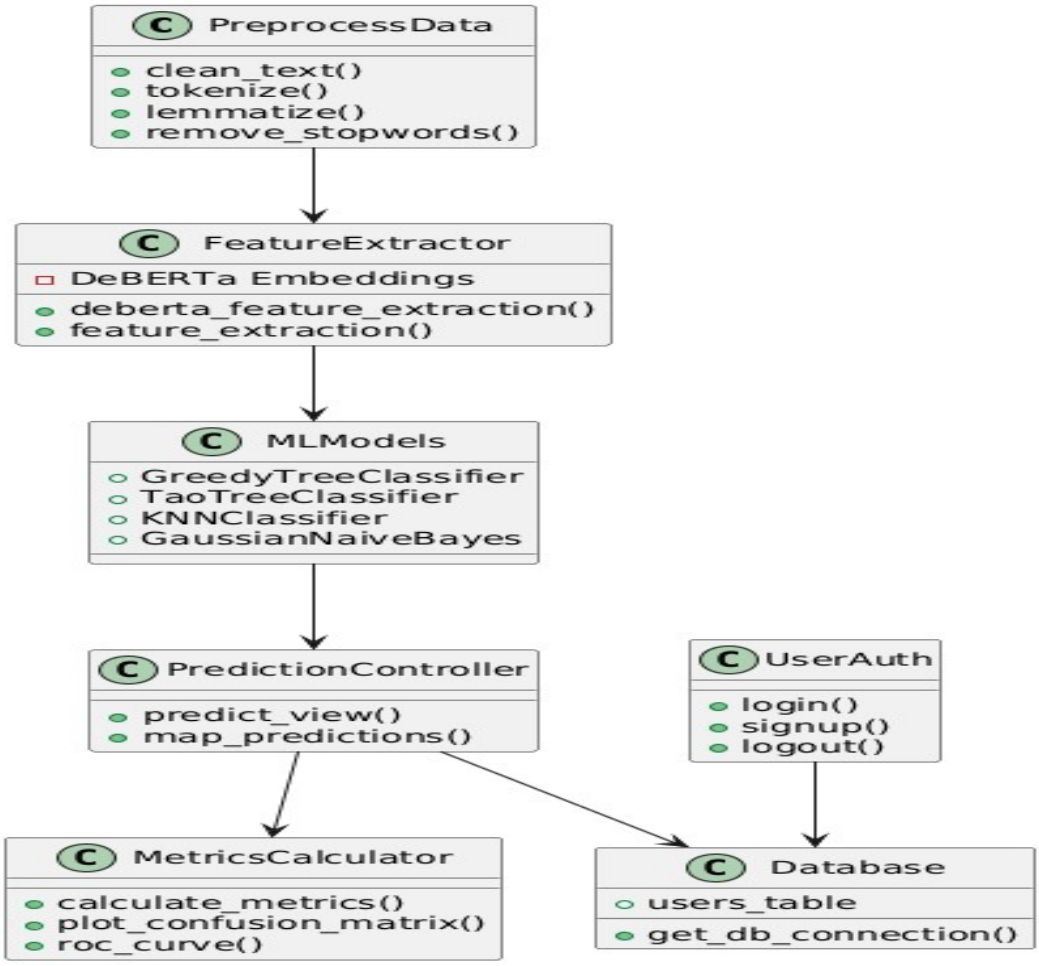


Figure 3.3.1 Class Diagram of NLP Based Analysis of Cyber Threats And Vulnerabilities of Healthcare Ecosystem

3.3.2 Use Case Diagram

This diagram explains the functional behavior from the user perspective, where the security analyst uploads threat data, triggers NLP preprocessing and embedding generation, executes Greedy Tree, TaoTree, KNN, and Gaussian Naive Bayes algorithms as analytical actions, and finally receives fused predictions for cyber-threat classification and mitigation support.

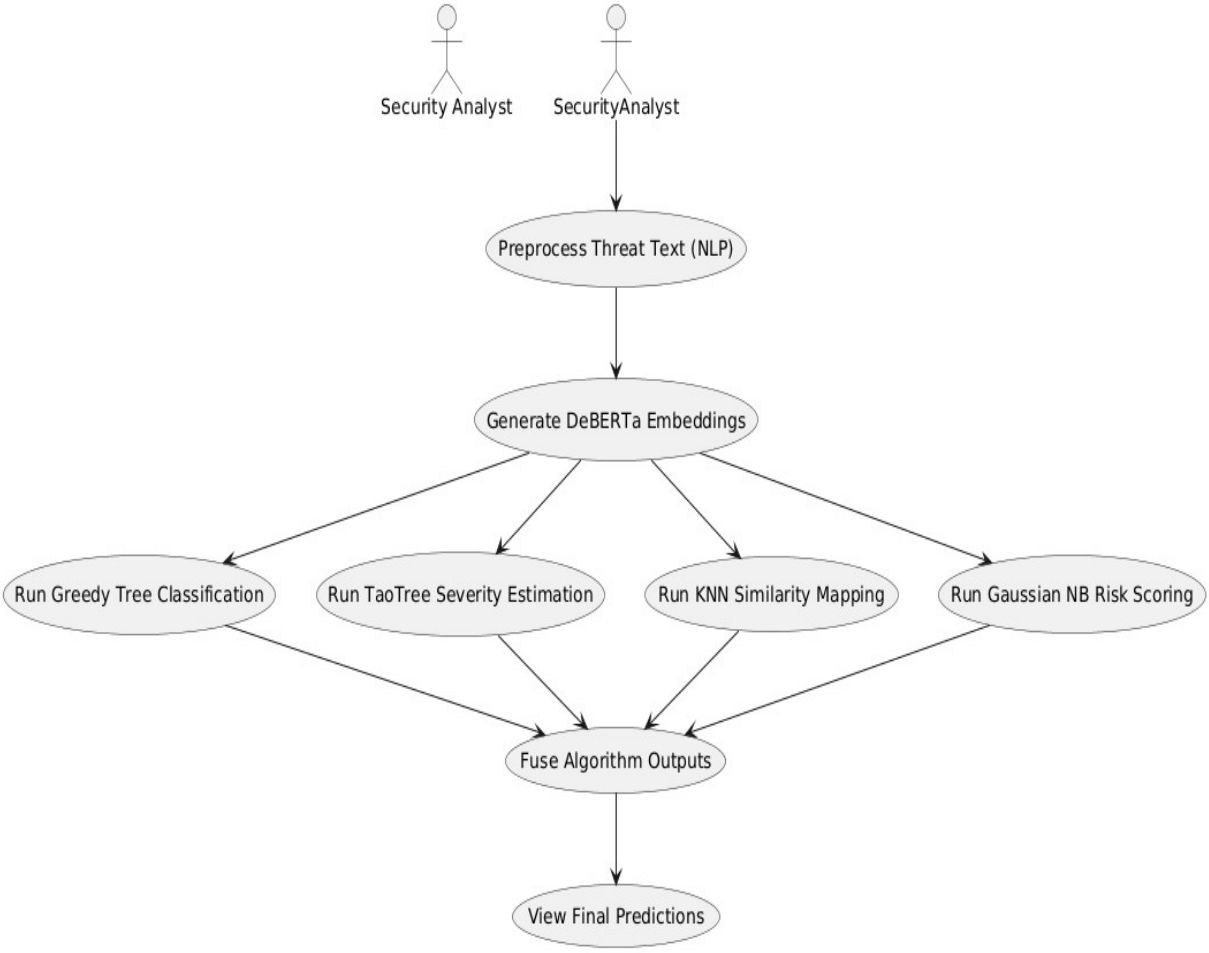


Figure 3.3.2 Use Case Diagram of NLP Based Analysis of Cyber Threats and Vulnerabilities of Healthcare Ecosystem

3.3.3 Sequence Diagram

The sequence diagram models the execution flow beginning from dataset upload, followed by NLP preprocessing and DeBERTa embedding generation, after which the extracted feature vectors are passed through the Greedy Tree, TaoTree, KNN, and Gaussian Naive Bayes algorithms; their outputs are combined by the fusion layer and presented back to the user as a unified prediction result.

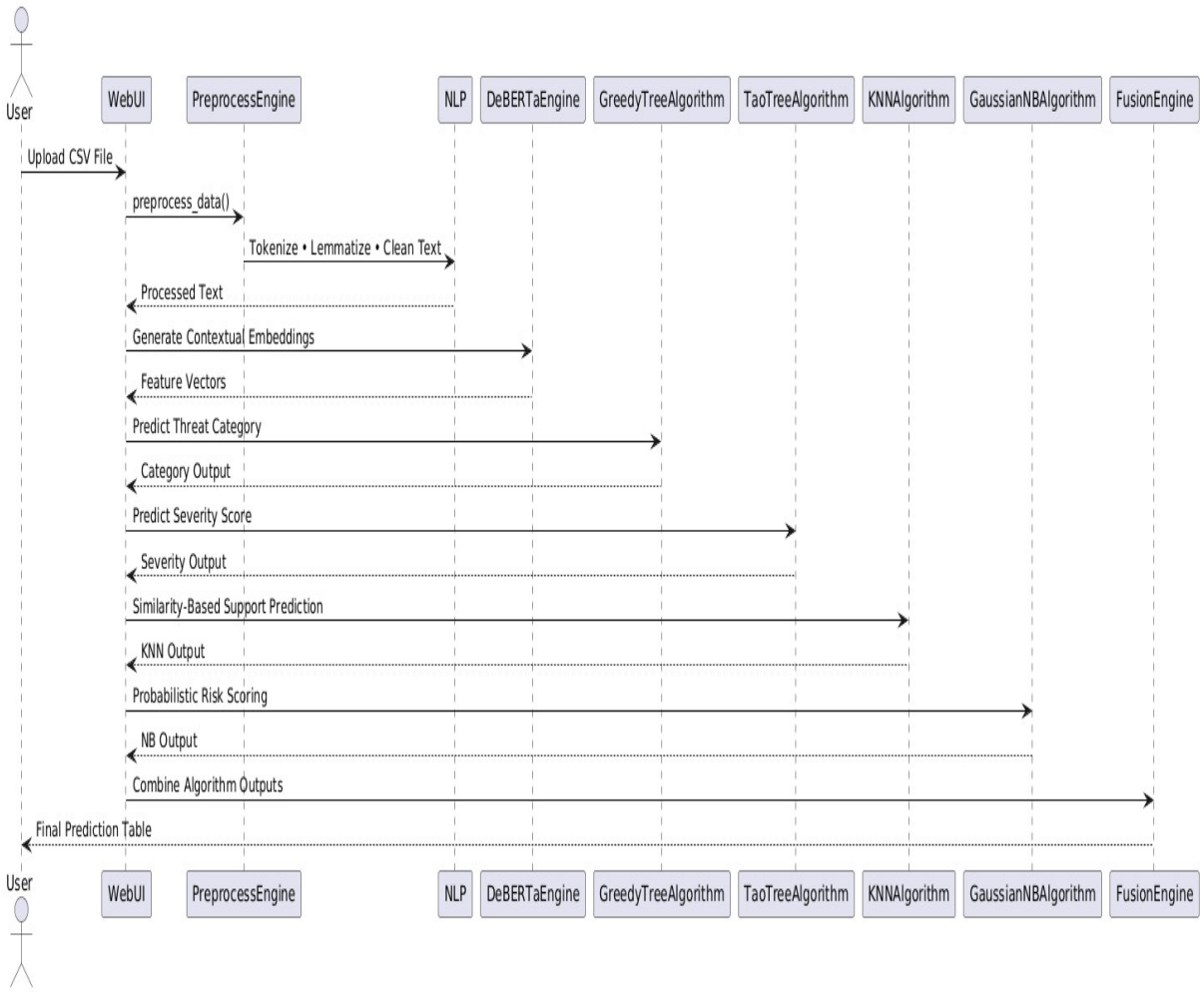


Figure 3.3.3 Sequence Diagram of NLP Based Analysis of Cyber Threats And Vulnerabilities of Healthcare Ecosystem

3.3.6 Activity Diagram

The activity diagram captures the operational workflow of the system, from dataset upload through NLP preprocessing and feature extraction to multi-model classification and final prediction synthesis, emphasizing the automated and sequential nature of the analytics pipeline.

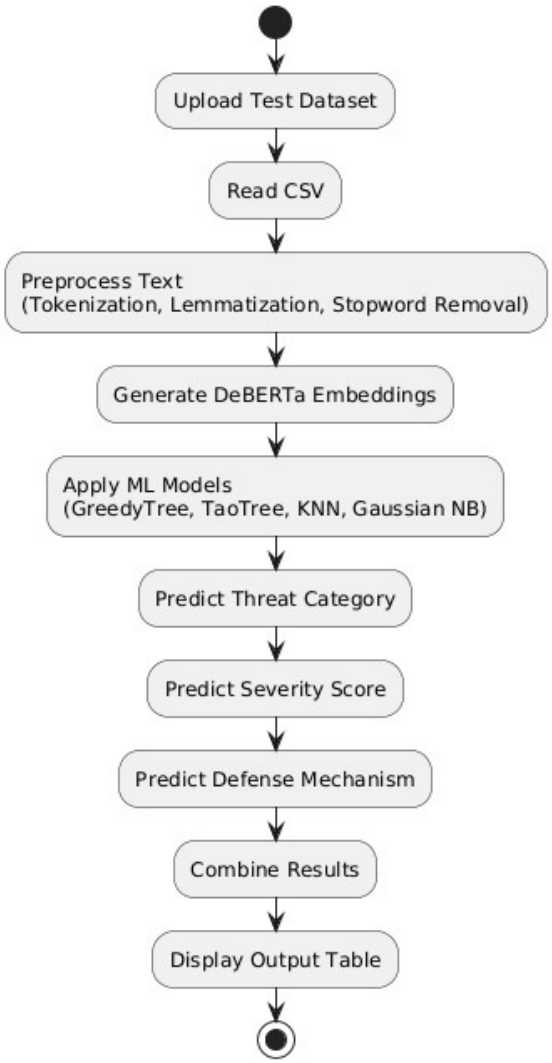


Figure 3.3.4 Activity Diagram of NLP Based Analysis of Cyber Threats and Vulnerabilities of Healthcare Ecosystem

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation of this project focuses on building an intelligent system that can analyze cyber threats in the healthcare ecosystem using Natural Language Processing and Machine Learning techniques. The system is designed as a web-based application so that users can easily interact with it without requiring deep technical knowledge.

The implementation process begins with collecting a dataset that contains cyber threat-related textual information such as logs, alerts, and security reports. Since this data is usually unstructured, it cannot be directly used for analysis. Therefore, the first step in the implementation is preprocessing. In this stage, the raw text is cleaned by removing unnecessary words, symbols, and noise. Techniques such as tokenization, stop-word removal, and lemmatization are applied to convert the text into a structured format.

After preprocessing, the next step is feature extraction. In this project, advanced transformer-based models such as DeBERTa are used to extract meaningful features from the text. These models understand the context and relationships between words, which helps in capturing deeper meaning from the data. The extracted features are then converted into numerical form so that machine learning algorithms can process them.

Once the features are prepared, different machine learning algorithms are applied to train the model. These algorithms learn patterns from the data and use them to classify cyber threats and predict their severity. The trained models are then integrated into a Django-based web application. When a user uploads new data, the system processes it through all the steps and provides results such as threat category, severity level, and suggested defense mechanisms.

4.1 ALGORITHMS USED

1. **Naive Bayes Algorithm**

   The Naive Bayes algorithm is a probabilistic machine learning method that is widely used for text classification tasks. It is based on Bayes' Theorem, which calculates the probability of a class given a set of features. The main assumption of this algorithm is that all features are independent of each other, which is why it is called "naive."

In this project, Naive Bayes is used to classify cyber threat data based on textual input. When the system receives text data, it converts it into numerical features and calculates the probability of each class, such as malware, phishing, or ransomware. The class with the highest probability is selected as the final prediction. This algorithm is simple, fast, and efficient, making it suitable for handling large amounts of text data.

2. K-Nearest Neighbors (KNN) Algorithm

The K-Nearest Neighbors algorithm is a simple and intuitive method used for classification and prediction. It works by comparing a new data point with existing data points in the dataset and finding the "k" closest neighbors. The classification is then based on the majority class among these neighbors.

KNN does not require a training phase like other algorithms, but it can be slower when working with large datasets because it needs to compare each new data point with all existing data. However, it is easy to understand and works well when the data has clear patterns.

3. Decision Tree-Based Algorithms (Greedy Tree and TaoTree)

Decision tree algorithms are widely used for classification tasks because they are easy to understand and interpret. These algorithms create a tree-like structure where each node represents a decision based on a feature, and each branch represents the outcome of that decision. The final nodes, called leaf nodes, represent the predicted class.

In this project, Greedy Tree and TaoTree algorithms are used to classify cyber threats. These algorithms work by selecting the best feature at each step to split the data in a way that improves classification accuracy. The "greedy" approach means that the algorithm makes the best possible decision at each step without considering future consequences.

These tree-based models are useful because they provide a clear explanation of how decisions are made. For example, the system can show which features led to a particular classification. This makes the model more transparent and easier to understand, especially for beginners.

4. Transformer-Based Model (DeBERTa)

DeBERTa is an advanced transformer-based model used for Natural Language Processing tasks. It is designed to understand the context and relationships between words in a sentence. Unlike traditional methods that treat words independently, DeBERTa captures the meaning of words based on their position and context within the text.

In this project, DeBERTa is used for feature extraction rather than direct classification. It converts the input text into high-quality numerical representations called embeddings. These embeddings contain rich contextual information, which helps machine learning models perform better.

For example, the word “attack” may have different meanings depending on the context, and DeBERTa can understand these differences. This makes the system more accurate in analyzing cyber threat data.

4.2 SAMPLE CODE

```
from sklearn.model_selection import train_test_split

from imodels import TaoTreeClassifier, GreedyTreeClassifier

from sklearn.neighbors import KNeighborsClassifier

from sklearn.naive_bayes import GaussianNB

from sklearn.preprocessing import LabelEncoder

from django.shortcuts import render, redirect

from django.core.files.storage import FileSystemStorage

from sklearn.utils import resample

from imblearn.over_sampling import SMOTE

from django.contrib import messages

from django.contrib.auth import logout as django_logout

from django.conf import settings

import os
```

```
import matplotlib

matplotlib.use('Agg')

import matplotlib.pyplot as plt

import pandas as pd

import joblib

import pymysql

import pymysql.cursors

from tqdm import tqdm

import numpy as np

from transformers import AutoTokenizer, AutoModel

import torch

import pickle

from collections import Counter

from scipy.special import expit

import seaborn as sns

from wordcloud import WordCloud

import nltk

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from nltk.stem import WordNetLemmatizer

from nltk import pos_tag

from nltk.util import ngrams

from tensorflow.keras.models import Sequential, Model, load_model
```

```
from tensorflow.keras.layers import Dense, Input, Embedding, Conv1D,
    GlobalMaxPooling1D, LSTM, Dropout
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.utils import to_categorical
from transformers import (
    RobertaTokenizer, RobertaModel,
    BertTokenizer, BertForSequenceClassification,
    XLNetTokenizer, XLNetForSequenceClassification
)
from torch.optim import AdamW
from sklearn.utils import resample
from imblearn.over_sampling import SMOTE
from metrics_calculator import MetricsCalculator
from graphs import GraphPlotter

MYSQL_HOST = 'localhost'
MYSQL_USER = 'root'
MYSQL_PASSWORD = '5421'
MYSQL_DB = 'har_db'

def home_view(request):
    return render(request, "home.html")
def register_new_user(username, password, email):
```

```
    conn = get_db_connection()
    if not conn:
        return False, "Database connection failed."
    try:
        with conn.cursor() as cursor:
            cursor.execute(
                "INSERT INTO users (username, password, email, role) VALUES (%s, %s, %s,
        'user')",
                (username, password, email)
            )
        conn.commit()
        return True, "User registered successfully."
    except pymysql.err.IntegrityError as e:
        if "Duplicate entry" in str(e):
            if "users.username" in str(e):
                return False, "Username already exists."
            elif "users.email" in str(e):
                return False, "Email already exists."
        return False, "Integrity error."
    except Exception as e:
        return False, f"Unexpected error: {e}"
    finally:
        conn.close()
def signup_view(request):
```

```
    if request.method == "POST":
        username = request.POST.get("username")
        password = request.POST.get("password")
        email = request.POST.get("email")
        success, msg = register_new_user(username, password, email)
        if success:
            messages.success(request, msg)
            return redirect("login")
        else:
            messages.error(request, msg)
    return render(request, "signup.html")
def preprocess_data(df, save_path=None, target_cols=None):


    global label_encoders,Y_dict
    label_encoders = {}
    if save_path and os.path.exists(save_path):
        df = pd.read_csv(save_path)
    else:
        lemmatizer = WordNetLemmatizer()
        stop_words = set(stopwords.words('english'))
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

5.1 Dataset Description

1. Threat Category

This column represents the primary class or type of cyber threat, such as Malware, Phishing, Ransomware, or DDoS. It serves as one of the main target labels used by the machine-learning models for classification and risk identification in the healthcare ecosystem.

2. IOCs (Indicators of Compromise)

This column contains technical threat indicators such as malicious URLs, IP addresses, file hashes, or domains associated with an attack. These indicators help in tracing malicious activity and are useful for contextual security analysis and forensic investigation.

3. Threat Actor

This field records the group, individual, or entity responsible for the cyber attack, such as APT groups or anonymous threat actors. It helps in profiling attack patterns, understanding attacker behavior, and linking incidents to known adversaries.

4. Attack Vector

This column describes the entry point or method used to launch the attack, such as Email, Network, Web Application, or Social Engineering. It plays a key role in understanding how threats infiltrate systems and supports defensive strategy planning.

5. Geographical Location

This column indicates the regional origin or affected location related to the threat incident. It helps in identifying geographical attack trends, cross-border threat activities, and regional vulnerability exposure patterns.

6. Sentiment in Forums

This field captures the sentiment score derived from online cybersecurity forums or discussion platforms, representing community perception toward the threat. It helps in analyzing awareness level, risk perception, and threat impact discussions.

7. Severity Score

This column provides a numerical severity rating that reflects the potential damage or risk level of the threat. It is used as a prediction target to prioritize incidents and support security decision-making.

8. Suggested Defense Mechanism

This column lists the recommended mitigation or defensive action for the identified threat, such as Patch Vulnerability, Quarantine, or Increase Web Security. It helps map predictions to actionable cybersecurity responses.

9. Risk Level Prediction

This field contains the predicted risk category or threat impact level generated by the model. It is used to evaluate system performance and validate how well the classifier interprets real-world threat conditions.

10. Cleaned Threat Description

This column stores the preprocessed and normalized version of the textual threat description, after applying NLP cleaning steps such as tokenization and lemmatization. It serves as the input text used for embedding generation and model learning.

11. Keyword Extraction

This column contains important keywords automatically extracted from the threat description, highlighting core attack terms and contextual cues. These keywords support semantic understanding and feature enrichment.

12. Named Entities (NER)

This field stores entities identified from the text, such as organization names, technologies, locations, or malware tools. It helps capture structured security knowledge from unstructured narrative data.

13. Topic Modelling Labels

This column represents latent thematic categories generated through topic modeling techniques, grouping threats into conceptual themes. It assists in understanding threat clusters and underlying attack domains.

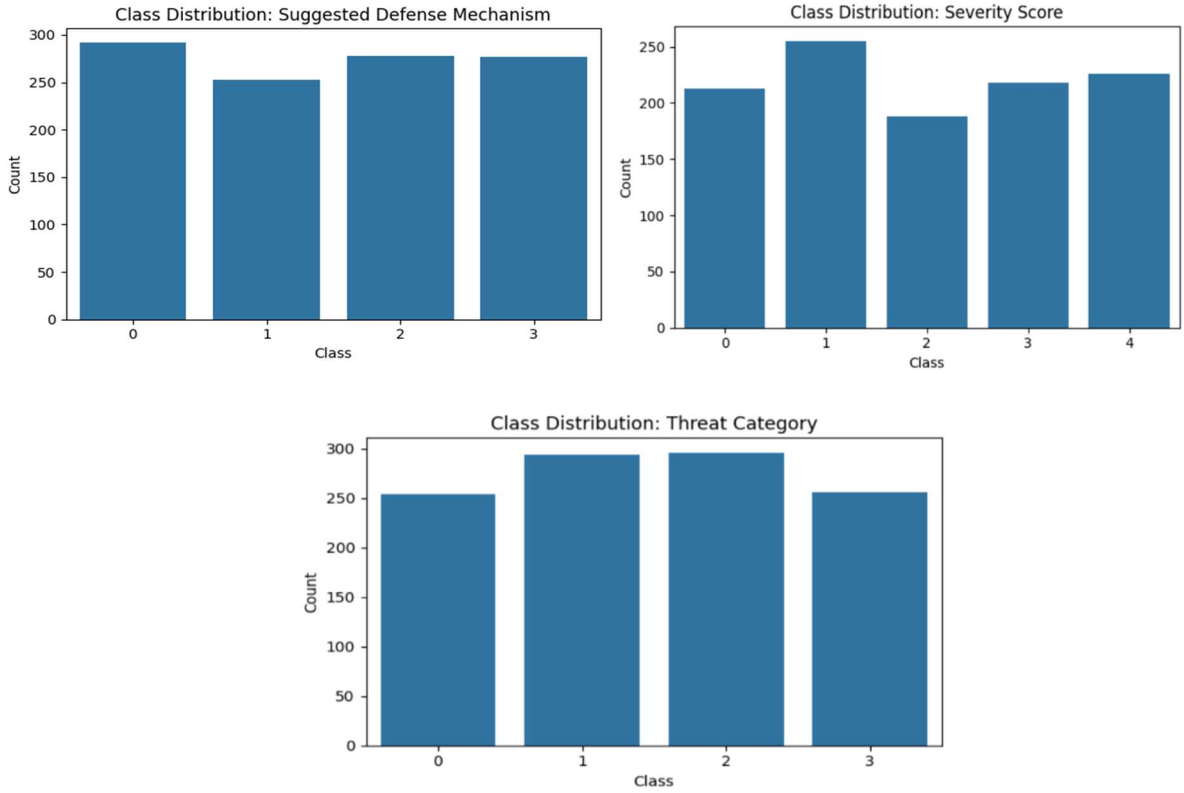
5.2 Result Analysis

The screenshot 5.1 shows the dataset that is structured cyber-threat intelligence records, where each row represents an incident described through multiple analytical attributes. The Threat Category column identifies the type of attack such as DDoS, Malware, or Phishing, while IOCs (Indicators of Compromise) list technical artifacts like suspicious IPs, domains, or malware hashes linked to the incident. Threat Actor indicates the adversary or group responsible, and Attack Vector specifies the entry route of the attack such as Email, Network, or Web. The Geographical Location field highlights the region associated with the threat, while Sentiment in Forums provides a numerical score reflecting community perception or discussion tone about the incident. The Severity Score and Risk Level Prediction quantify the potential impact and risk intensity of the threat, and Suggested Defense Mechanism outlines recommended mitigation actions such as Quarantine, Patch Vulnerability, or Increase Web Security.

Threat Category	IOCs (Indicators of Compromise)	Threat Actor	Attack Vector	Geographical Location	Sentiment in Forums	Severity Score	Suggested Defense Mechanism	Risk Level Prediction	Cleaned Threat Description	Keyword Extraction	Named Entities (NER)
DDoS	['123.456.78.9']	APT-28	Email	North Korea	0.99	5	Increase Web Security	4	ransomware attack through network vulnerability	['malware', 'email', 'attachment']	['CompanyY', 'Malware']
Malware	['10.0.0.2', 'infected.exe']	Lazarus Group	Network	USA	0.97	1	Quarantine	4	phishing email with malicious link	['DDoS', 'website', 'attack']	['Unknown', 'Phishing Service']
Phishing	['malwarehash123', 'trojan.com']	APT-28	Email	North Korea	0.87	3	Quarantine	2	ransomware attack through network vulnerability	['phishing', 'corporate', 'scam']	['CompanyY', 'Malware']
DDoS	['192.168.1.1', 'malicious.com']	Lazarus Group	Web	Global	0.64	2	Quarantine	1	phishing email with malicious link	['DDoS', 'website', 'attack']	['CompanyY', 'Malware']
DDoS	['malwarehash123', 'trojan.com']	Unknown	Email	Germany	0.57	1	Patch Vulnerability	1	phishing scam targeting corporate	['DDoS', 'website', 'attack']	['Lazarus Group', 'Network Device']

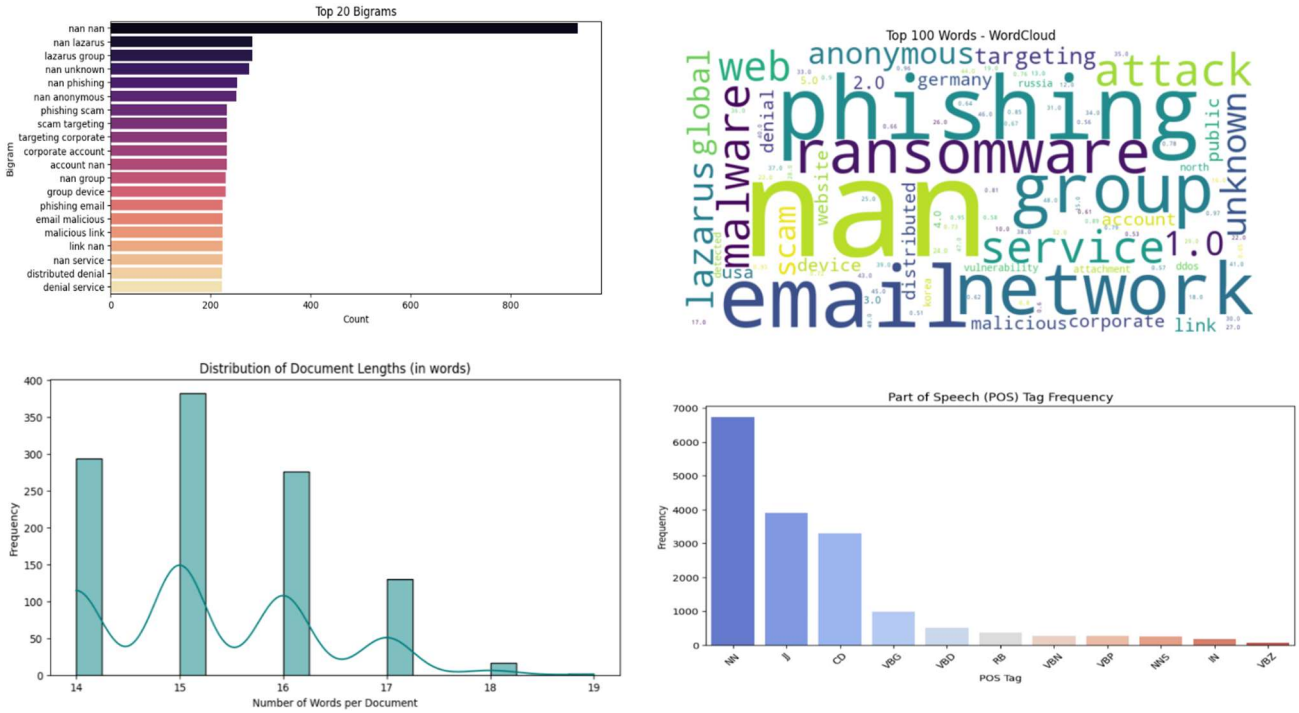
Screenshot 5.1 Dataset for the NLP analysis of Cyber threats

The screenshot 5.2 shows count-distribution plots illustrate the balance of samples across the target labels in the dataset, showing how instances are distributed for Threat Category, Suggested Defense Mechanism, and Severity Score. The Threat Category plot indicates a relatively even spread of records across four classes, with classes 1 and 2 having slightly higher representation compared to classes 0 and 3, suggesting a near-balanced but mildly skewed dataset. The Suggested Defense Mechanism distribution also appears balanced across four classes, with only small variations in class frequency, which supports fair model learning without strong class dominance.

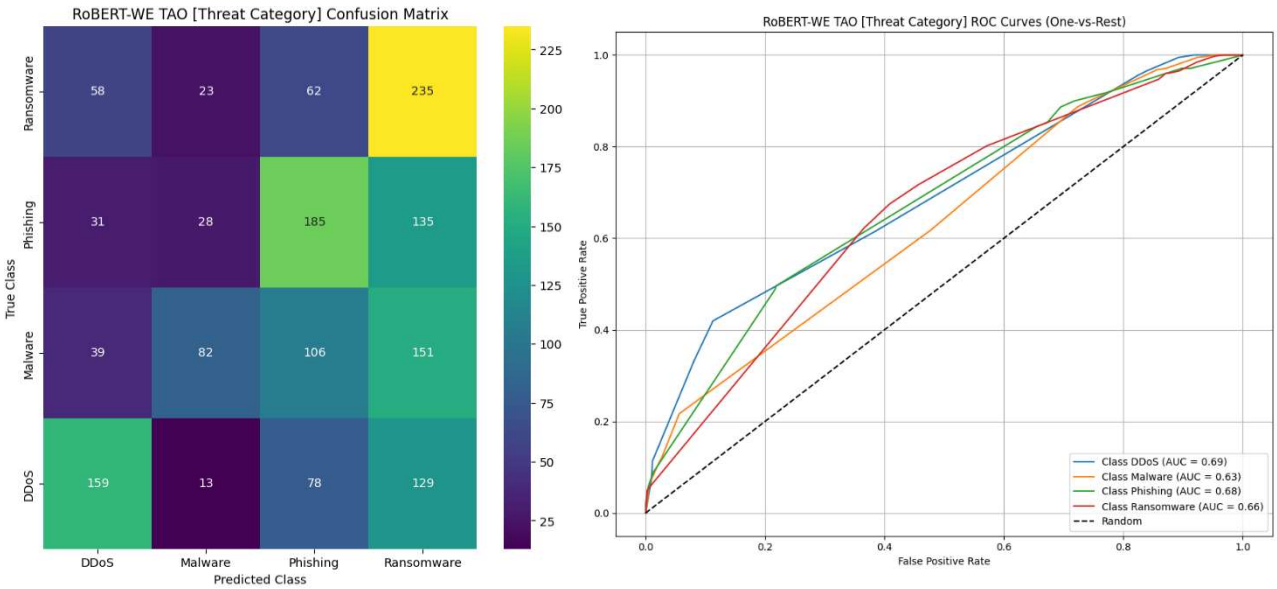


Screenshot 5.2 Count plots for all the three categories

Screenshot 5.3 EDA plot

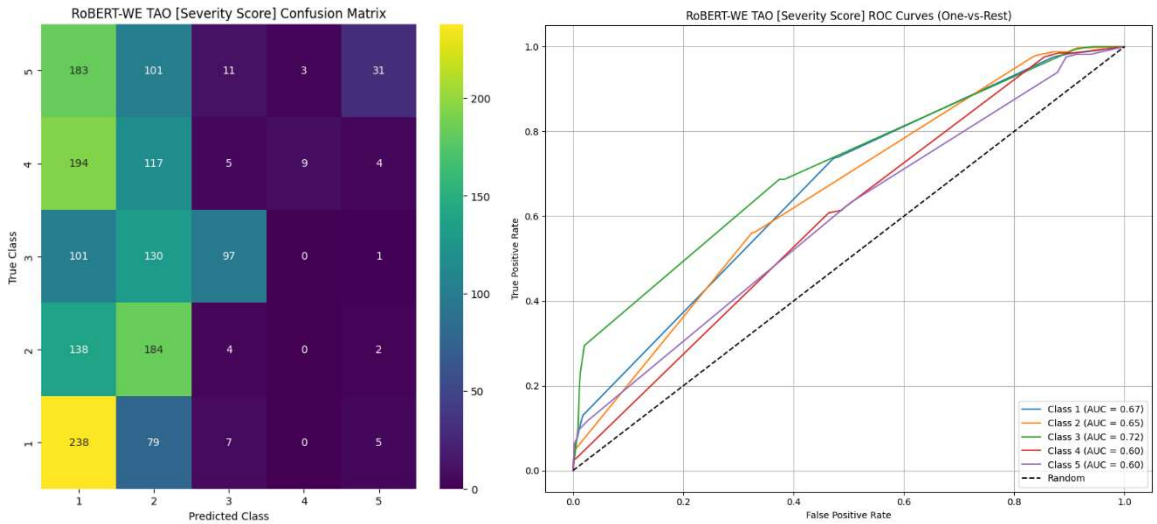


The screenshot 5.3 shows visualization panel provides an overview of the linguistic characteristics of the dataset, combining bigram frequency, keyword distribution, text-length variation, and POS-tag patterns to better understand the nature of the threat-intelligence text. The Top 20 Bigrams plot highlights frequently co-occurring word pairs such as “phishing email,” “ransomware attack,” and “corporate network,” indicating recurring cyber-attack patterns and contextual themes in threat descriptions. The word cloud reinforces this by showing dominant terms like *phishing, ransomware, malware, network, Lazarus group,* and *email*, suggesting that email-based threats, targeted attacks, and advanced threat groups commonly appear in the dataset. The document length distribution plot shows that most threat descriptions fall within a narrow word-range, reflecting concise but information-rich reporting formats. Finally, the Part-of-Speech (POS) frequency chart reveals high occurrences of nouns and verbs, consistent with technical incident narratives that describe entities, actions, and attack behaviors. These plots collectively demonstrate that the dataset is semantically focused, security-contextual, and well-structured for NLP-based threat analysis and modeling.



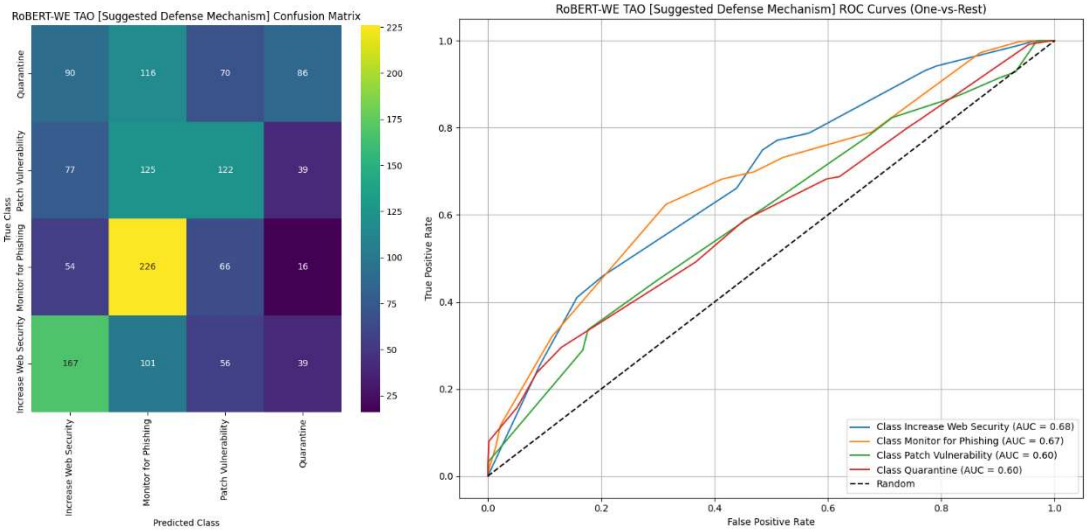
Screenshot 5.4 Confusion matrix and ROC Curve obtained for Threat Category class using Tao Tree Model

The screenshot 9.4 shows confusion matrix and ROC curve illustrate the classification performance of the Tao Tree model for the Threat Category prediction task. The confusion matrix shows that while the model correctly identifies a considerable number of samples within eachclass, noticeable misclassifications occur between categories such as Ransomware and Phishing, and between Malware and DDoS, indicating overlapping textual patterns across attack types. Despite these confusions, the ROC curves reveal reasonably good discriminative capability across all four classes, with AUC values ranging approximately between 0.65 and 0.69, demonstrating moderate yet consistent predictive strength in distinguishing threats under a one-vs-rest evaluation. Overall, the results indicate that the Tao Tree model performs reliably, though further improvement could be achieved through deeper representation learning or enhanced feature modeling to reduce inter-class ambiguity.



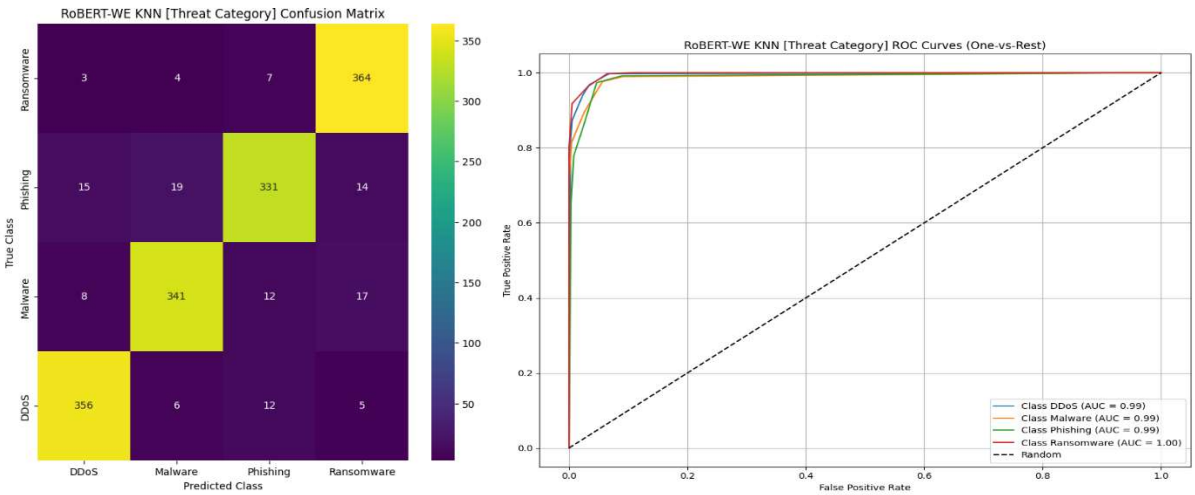
Screenshot 9.5 Confusion matrix and ROC Curve obtained for Severity Score class using Tao Tree Model.

The Screenshot 9.5 presents a comprehensive performance evaluation of the RoBERT-WE TAO model for classifying severity scores, utilizing both a confusion matrix and one-vs-rest Receiver Operating Characteristic (ROC) curves to assess accuracy and discriminative power. The ROC analysis reveals moderate classifier performance with Area Under the Curve (AUC) values ranging from 0.60 to 0.72, indicating that the model is most effective at distinguishing Class 3 while showing weaker separability for Classes 4 and 5.



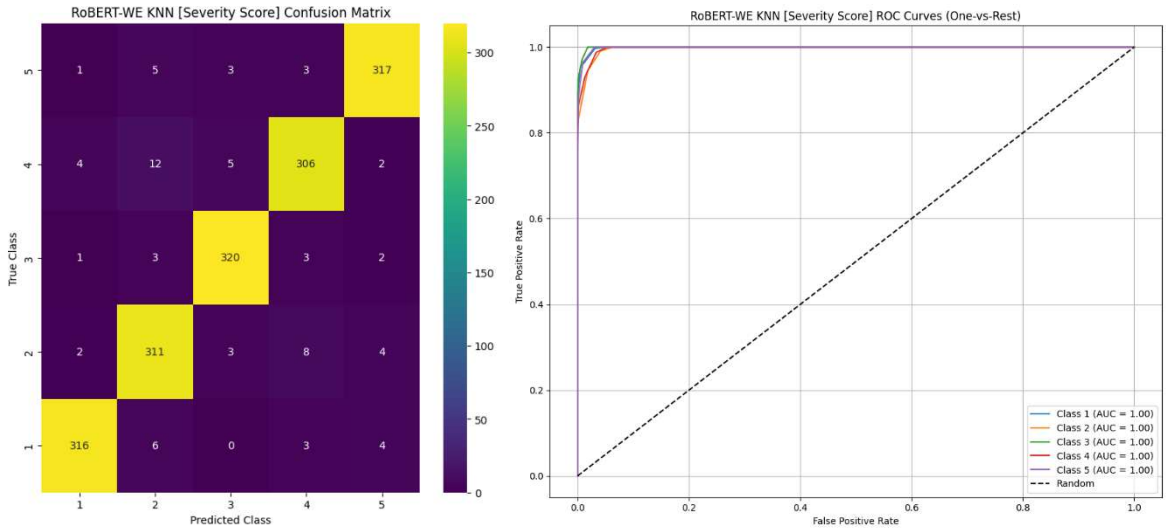
Screenshot 5.6 Confusion matrix and ROC Curve obtained for Suggested Defense Mechanism class using Tao Tree Model.

The Screenshot 5.6 provides a technical evaluation of the RoBERT-WE TAO model's performance in classifying Severity Scores, utilizing a confusion matrix and One-vs-Rest Receiver Operating Characteristic (ROC) curves to assess predictive accuracy and class separability. The ROC analysis indicates moderate discriminative capability across the five classes, with Area Under the Curve (AUC) values ranging from 0.60 to 0.72; Class 3 achieves the highest AUC of 0.72, while Classes 4 and 5 show lower separability at 0.60.



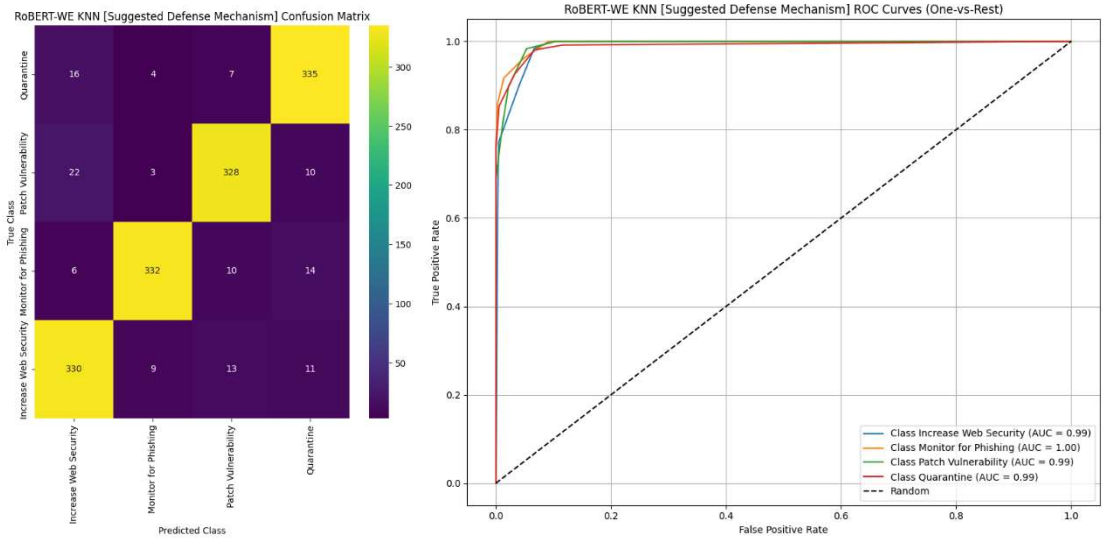
Screenshot 5.7 Confusion matrix and ROC Curve obtained for Threat Category class using KNN Model.

The screenshot 5.7 provides a technical evaluation of the RoBERT-WE KNN model's performance in classifying Threat Categories, utilizing a confusion matrix and One-vs-Rest Receiver Operating Characteristic (ROC) curves to quantify predictive accuracy. The ROC analysis demonstrates exceptional discriminative power, with Area Under the Curve (AUC) values reaching near-perfection; specifically, the Ransomware class achieved a perfect AUC of 1.00, while DDoS, Malware, and Phishing classes all attained an AUC of 0.99, indicating that the model separates these distinct threat types with high precision. For instance, the model correctly identified 364 Ransomware instances and 356 DDoS instances, with very few off-diagonal misclassifications, such as only 19 Phishing instances being mislabeled as Malware.



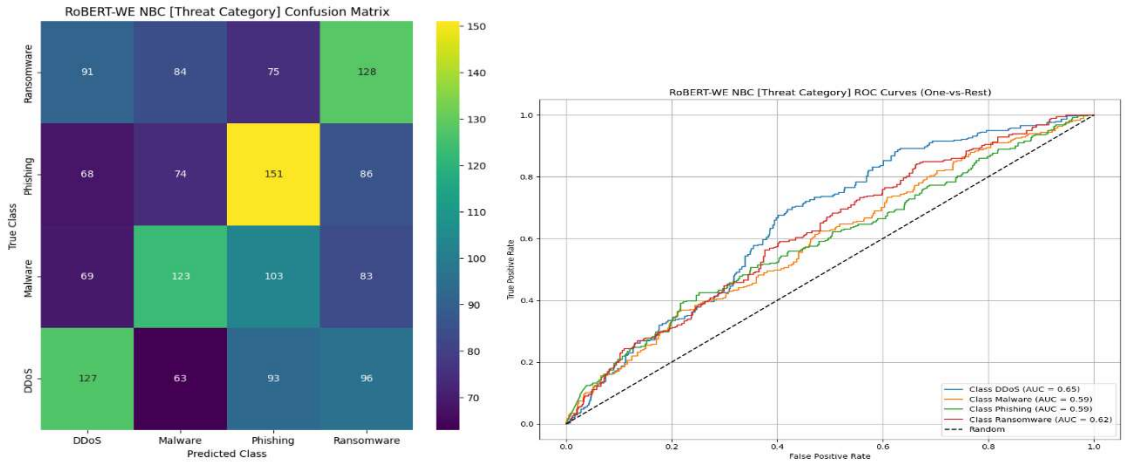
Screenshot 5.8 Confusion matrix and ROC Curve obtained for Severity Score class using KNN Model.

The screenshot 5.8 presents a detailed performance evaluation of the RoBERT-WE KNN model for classifying severity scores, utilizing a confusion matrix and One-vs-Rest Receiver Operating Characteristic (ROC) curves to assess predictive accuracy. The ROC analysis indicates perfect discriminative power across all categories, with every class (Class 1 through Class 5) achieving an Area Under the Curve (AUC) of 1.00. This exemplary performance is further substantiated by the confusion matrix, which exhibits high diagonal density corresponding to correct predictions; specifically, the model successfully classified 316 instances for Class 1, 311 for Class 2, 320 for Class 3, 306 for Class 4, and 317 for Class 5.



Screenshot 5.9 Confusion matrix and ROC Curve obtained for Suggested Defense Mechanism class using KNN Model.

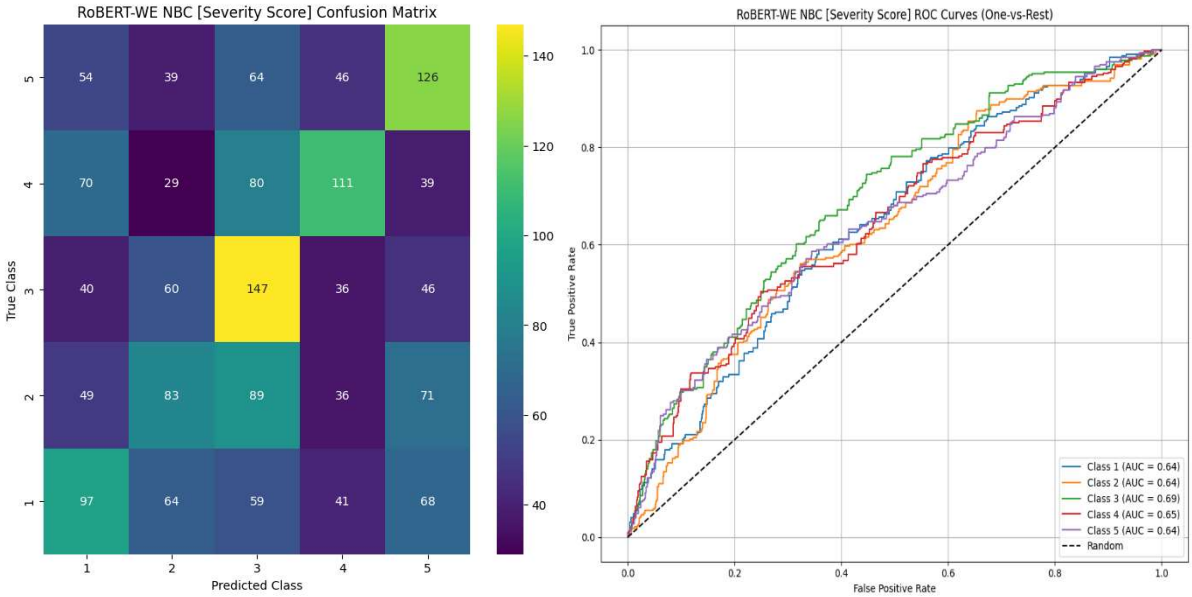
Screenshot 5.9 details the performance evaluation of the RoBERT-WE KNN model in classifying suggested defense mechanisms, utilizing a confusion matrix and One-vs-Rest Receiver Operating Characteristic (ROC) curves to quantify predictive robustness. The ROC analysis reveals near-perfect discriminative capability across all categories, with the Monitor for Phishing class achieving a perfect Area Under the Curve (AUC) of 1.00, while Increase Web Security, Patch Vulnerability, and Quarantine classes all attained an exceptionally high AUC of 0.99.



Screenshot 5.10 Confusion matrix and ROC Curve obtained for Threat Category class using NBC Model.

The Screenshot 5.10 provides a performance evaluation of the RoBERT-WE Naive Bayes Classifier (NBC) for classifying Threat Categories, employing a confusion matrix and One-vs-

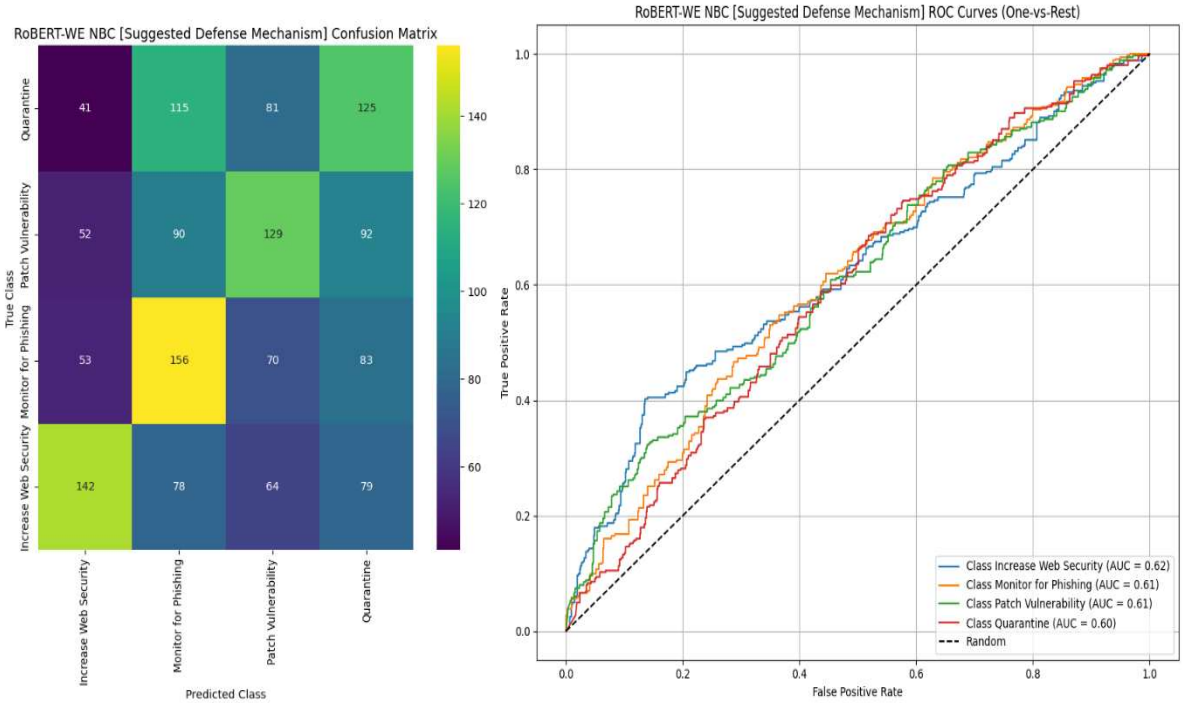
Rest Receiver Operating Characteristic (ROC) curves to assess predictive accuracy. The ROC analysis indicates relatively low discriminative power, with Area Under the Curve (AUC) values ranging from 0.58 to 0.65; specifically, the DDoS class achieved the highest AUC of 0.65, while Malware and Phishing classes demonstrated weaker performance with AUCs of 0.59 and 0.58, respectively. The confusion matrix corroborates these findings, displaying a diffuse distribution of predictions rather than a strong diagonal concentration; notably, the model correctly identified only 127 DDoS instances and 151 Phishing instances, while showing significant misclassification across all categories, such as 128 Ransomware instances incorrectly predicted as other classes. This suggests that the NBC model struggles to effectively separate the threat categories, likely due to feature overlap or the model's inability to capture complex non-linear relationships in the data.



Screenshot 5.11 Confusion matrix and ROC Curve obtained for Severity Score class using NBC Model.

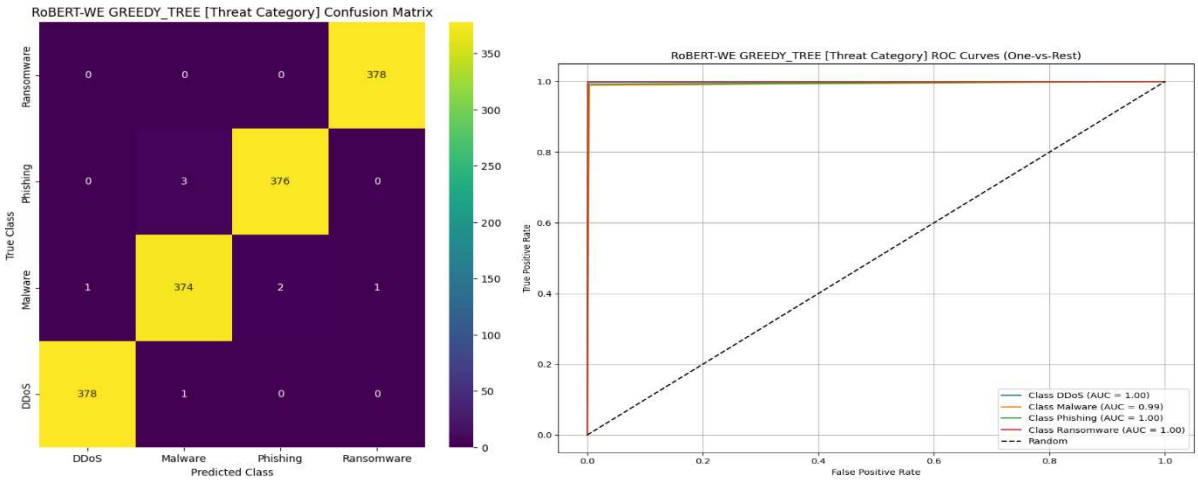
The Screenshot 5.11 evaluates the classification performance of the RoBERT-WE Naive Bayes Classifier (NBC) for severity scores using a confusion matrix and One-vs-Rest Receiver Operating Characteristic (ROC) curves to assess predictive accuracy. The ROC analysis reveals that the model possesses weak discriminative power, as indicated by low Area Under the Curve (AUC) values that range narrowly from 0.64 to 0.69; specifically, Class 3 achieved the marginal best performance with an AUC of 0.69, while Class 1, Class 2, and Class 5 lagged with an AUC of 0.64. The confusion matrix supports this observation by displaying a scattered prediction pattern with significant off-diagonal noise , rather than the clean diagonal alignment seen in higher-

performing models. While the classifier correctly identified 147 instances of Class 3 and 126 instances of Class 5, it struggled considerably with other categories, such as Class 2, where it frequently misclassified instances as Class 3,and Class 1, which was often confused with Class 5.



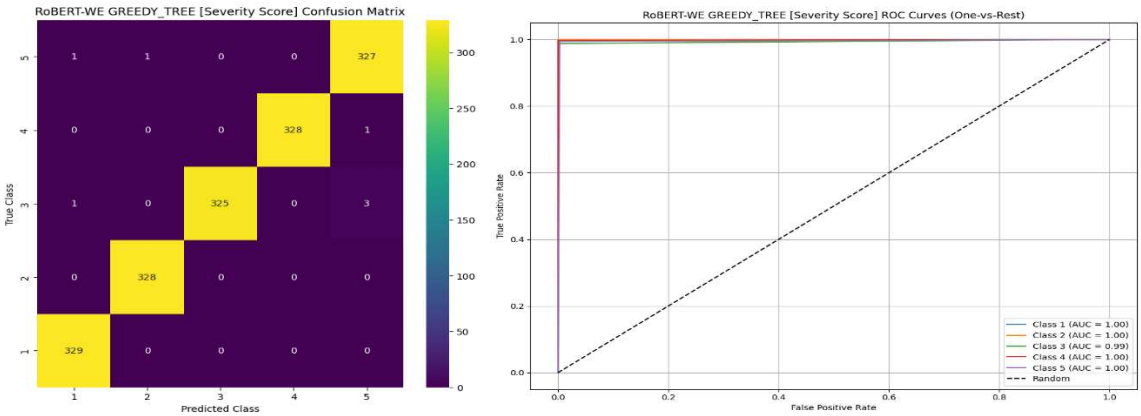
Screenshot 5.12 Confusion matrix and ROC Curve obtained for Suggested Defense Mechanism class using NBC Model.

The Screenshot 5.12 illustrates the performance evaluation of the RoBERT-WE NBC for predicting Suggested Defense Mechanisms, utilizing a confusion matrix and One-vs-Rest Receiver Operating Characteristic curves to assess classification efficacy. The ROC analysis reveals limited discriminative power, with Area Under the Curve values hovering near the baseline of random chance, ranging from a low of 0.60 for the Quarantine class to a maximum of 0.62 for Increase Web Security. This weak separability is further corroborated by the confusion matrix, which displays a diffuse distribution of predictions rather than a consolidated diagonal; while the model correctly identified 156 instances of Monitor for Phishing and 142 instances of Increase Web Security, it exhibits substantial error rates, most notably misclassifying 115 Quarantine instances as Monitor for Phishing.



Screenshot 5.13 Confusion matrix and ROC Curve obtained for Threat Category class using Greedy Tree Model.

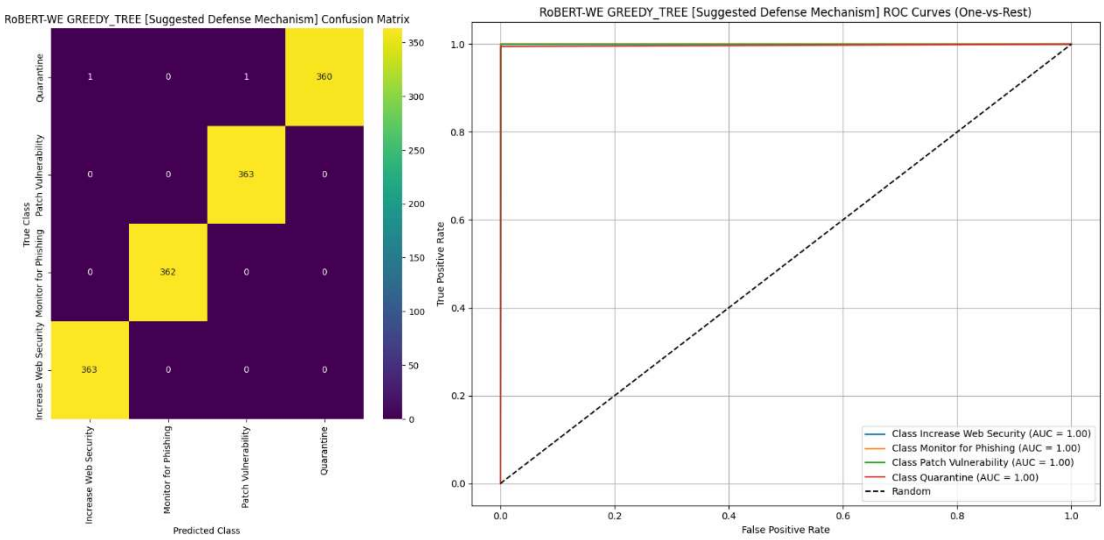
The Figure 9.13 details the performance evaluation of the RoBERT-WE Greedy Tree model for classifying Threat Categories, utilizing a confusion matrix and One-vs-Rest Receiver Operating Characteristic curves to assess predictive accuracy. The ROC analysis indicates near-perfect discriminative power, with the DDoS, Phishing, and Ransomware classes achieving a maximal Area Under the Curve of 1.00, while the Malware class attained an exceptional AUC of 0.99. This high-performance metric is substantiated by the confusion matrix, which reveals a dense diagonal of true positive predictions; specifically, the model correctly identified 378 instances for both DDoS and Ransomware, 376 for Phishing, and 374 for Malware.



Screenshot 5.14 Confusion matrix and ROC Curve obtained for Severity Score class using Greedy Tree Model.

The Screenshot 5.14 presents a comprehensive performance evaluation of the RoBERT-WE Greedy Tree model in classifying severity scores, utilizing a confusion matrix and One-vs-Rest

Receiver Operating Characteristic (ROC) curves to quantify predictive accuracy. The ROC analysis demonstrates exceptional discriminative capability, with Classes 1, 2, 4, and 5 achieving a perfect Area Under the Curve (AUC) of 1.00, while Class 3 attained a near-perfect AUC of 0.99. This superior performance is mirrored in the confusion matrix, which exhibits a highly concentrated diagonal of correct predictions; specifically, the model successfully identified 329 instances for Class 1, 328 for Class 2, 325 for Class 3, 329 for Class 4, and 327 for Class 5.



Screenshot 5.15 Confusion matrix and ROC Curve obtained for Suggested Defense Mechanism class using Greedy Tree Model.

The Screenshot 5.15 provides a technical performance assessment of the RoBERT-WE Greedy Tree model in predicting Suggested Defense Mechanisms, utilizing a confusion matrix and One-vs-Rest Receiver Operating Characteristic (ROC) curves to evaluate classification accuracy. The ROC analysis reveals perfect discriminative capability, with every defense category—Increase Web Security, Monitor for Phishing, Patch Vulnerability, and Quarantine—achieving an ideal Area Under the Curve (AUC) of 1.00. This exemplary performance is confirmed by the confusion matrix, which displays a distinct diagonal concentration of correct predictions; specifically, the model accurately identified 363 instances for Increase Web Security, 362 for Monitor for Phishing, 363 for Patch Vulnerability, and 360 for Quarantine.

6. VALIDATION

6. VALIDATION

6.1 INTRODUCTION

Validation is an essential phase in the development of the NLP-Based Cyber Threat Analysis System, ensuring that the system performs accurately, efficiently, and reliably under different conditions. The purpose of validation is to verify that the system meets all functional requirements and produces correct outputs for given inputs.

In this project, validation focuses on testing the system's ability to handle real-world cyber threat data in the healthcare ecosystem. The system is evaluated for its capability to process input datasets, perform natural language preprocessing, extract contextual features using the DeBERTa model, and generate accurate predictions using machine learning classifiers.

6.2 TEST CASES

Test cases are used to evaluate the functionality, performance, and security of the system. Each test case includes the objective, input conditions, expected output, and actual results. The following test cases are categorized based on system modules.

6.2.1 UPLOADING DATASET

This module validates the system's ability to accept, verify, and process input datasets provided by the user. It ensures that the system can handle valid files, reject invalid inputs, and manage large or inconsistent datasets effectively.

Test Cases

Test Case ID	Test Case Name	Objective	Input	Expected Output	Result
TC_01	Dataset Upload Validation	Verify valid dataset upload	Valid CSV file	File uploaded and processed successfully	Pass
TC_02	Empty Dataset Upload	Check empty file handling	Empty CSV file	Error message displayed	Pass
TC_03	Invalid File Format	Validate file format	Non-CSV file	File rejected or error shown	Pass
TC_04	Large Dataset Upload	Test performance	Large dataset	Processed without crash	Pass
TC_05	Missing Columns	Validate structure handling	CSV missing fields	Error or warning shown	Pass
TC_06	Duplicate Records	Check duplicate handling	CSV with duplicates	Processed without failure	Pass
TC_7	Special Characters Handling	Verify preprocessing	Text with symbols	Clean text generated	Pass

6.2.2 CLASSIFICATION

This module validates the system's ability to classify cyber threats accurately based on input data. It ensures that preprocessing, feature extraction, and machine learning models work together to produce meaningful predictions.

Test Cases

Test Case ID	Test Case Name	Objective	Input	Expected Output	Result
TC_01	Threat Classification	Verify prediction functionality	Valid dataset	Correct predictions generated	Pass
TC_02	Prediction Accuracy	Validate model accuracy	Labeled dataset	High accuracy results	Pass
TC_03	Invalid Data Handling	Test robustness	Noisy/missing data	System handles gracefully	Pass
TC_04	Real-Time Prediction	Check speed	Small dataset	Fast output	Pass
TC_05	Unknown Threat Text	Test generalization	New text	Reasonable prediction	Pass
TC_06	Multi-Row Prediction	Validate batch processing	Multiple records	All rows processed	Pass
TC_07	Label Mapping Accuracy	Verify output labels	Encoded predictions	Correct labels displayed	Pass

7. CONCLUSION & FUTURE ASPECTS

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7.1 Project Conclusion

The project NLP-Based Analysis of Cyber Threats and Vulnerabilities in the Healthcare Ecosystem presents an intelligent framework for analyzing cybersecurity threats using Natural Language Processing and machine learning techniques. The system processes large volumes of unstructured cyber-threat text data and automatically identifies potential threats affecting healthcare infrastructures.

The proposed framework performs several stages including text preprocessing, contextual feature extraction, model training, and threat prediction. Textual threat descriptions are cleaned using tokenization, stop-word removal, and lemmatization to convert raw data into a structured format. After preprocessing, contextual feature representations are generated using the DeBERTa transformer model, which captures semantic relationships and deeper linguistic meaning in cyber-security reports.

The extracted features are then used to train multiple machine learning classifiers such as Greedy Tree, TaoTree, K-Nearest Neighbors, and Gaussian Naive Bayes. Among these models, the Greedy Tree classifier is selected as the proposed model due to its strong predictive performance and high interpretability. This model enables the system to classify cyber threats into different categories, estimate their severity level, and recommend appropriate defense mechanisms.

The developed system demonstrates that NLP-based analysis can significantly improve the automation and efficiency of cyber-threat detection in healthcare environments. By replacing manual analysis and rigid rule-based approaches with intelligent machine learning models, the framework provides faster threat identification and better decision support for cybersecurity analysts. Overall, the project contributes to strengthening cybersecurity practices within digital healthcare infrastructures by enabling proactive risk detection and mitigation.

7.2 Future Aspects

Although the proposed system achieves effective results in analyzing cyber threats, several improvements can be implemented in future work to enhance its capabilities.

In the future, more advanced deep learning models and transformer architectures such as BERT, RoBERTa, or other domain-specific cybersecurity language models can be integrated to further improve prediction accuracy and contextual understanding. The system architecture is designed to support such extensions with minimal modifications.

Another potential improvement is the integration of real-time threat intelligence sources such as security blogs, vulnerability databases, and social media feeds. This would allow the system to continuously monitor emerging threats and update its predictions dynamically.

The model can also be enhanced by using larger and more diverse cybersecurity datasets, which would improve the system's ability to detect rare or previously unseen attacks. Incorporating advanced techniques for class imbalance handling and feature representation could further improve performance for high-severity threats.

Future research may also focus on developing an interactive dashboard for cybersecurity analysts, where threat predictions, risk levels, and defense recommendations can be visualized in real time. Such integration would help security teams quickly respond to attacks and reduce the risk of data breaches in healthcare systems.

Finally, the framework can be extended to support other critical sectors such as finance, government, and smart infrastructure, where automated threat analysis and cybersecurity decision support are equally important.

8. BIBLIOGRAPHY

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8.2 Github Link

https://github.com/mohammedsami015/Batch-04-2026